# Atmo-Ensemble
Ensemble ML Research Conclusion for Air Quality Multi-Classification

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March 2026

**Abstract**

# 1 Research Conclusions

## 1.1 PCA and Dataset Feature Extraction

*Does studying correlated environmental features through principal component analysis (PCA) and other feature extraction techniques improve the robustness and efficiency of ensemble classifiers, or are non-linear feature interactions found by the models alone?*

## 1.2 Feature Importance and Interactions (XAI)

XAI context

## 1.3 Ensemble Learning

*Does the variance-reduction of parallel ensembles offer better generalization on noisy sensor data compared to the bias-reduction of sequential ensembles?*

### 1.3.1 Overview of Evaluated Models

To address this task, we built different ensemble learning based models and evaluated them to classify air quality into four categories: Good, Moderate, Poor, and Hazardous. The main focus was getting a final model able to correctly predict pollution. The evaluated models include:

- **Voting Ensembles**: Hard voting (majority) and Soft voting (average probability) combining Logistic Regression, Random Forest, and Support Vector Classifier.
- **Parallel Ensembles (Variance-Reduction)**: Bagging (with replacement) and Pasting (without replacement) utilizing parallel training on data subsets.
- **Sequential Ensembles (Bias-Reduction)**: Boosting methods like AdaBoost and Gradient Boosting, which use sequential training to correct prior errors.

- **XGBoost**: A high-performance gradient boosting implementation featuring system optimization and regularization.
- **Stacking Ensemble**: A multi-layer ensemble using a final meta-learner to blend predictions.

Because of the class imbalances in the dataset, special importance was given to the Macro F1-score alongside Accuracy, Weighted F1-Score, and the Matthews Correlation Coefficient (MCC).

### 1.3.2 Parallel vs. Sequential Ensembles Performance

The parallel tree ensembles, Bagging and Pasting, demonstrated strong variance reduction, achieving Macro F1-scores of 0.9240 and 0.9204, respectively. Their out-of-bag (OOB) score of 0.9504 provided a reliable estimate of model performance. However, sequential ensembles, which focus on correcting the errors of prior models, ultimately outperformed them.

While the fine-tuned AdaBoost model reached a Macro F1 of 0.9209, Gradient Boosting achieved a higher Macro F1 of 0.9297, bypassing the parallel methods. Finally, XGBoost obtained the highest overall scores with an Accuracy of 0.9560, a Macro F1 of 0.9351, and an MCC of 0.9371.

### 1.3.3 Key Insights and Conclusion

Based on the gathered metrics, the bias-reduction approach of sequential ensembles offered better generalization on the sensor data compared to the variance-reduction of parallel ensembles. While bagging offered robustness and stability, the boosting methods, particularly XGBoost, provided the best predictive performance.

Despite the strong overall performance, a key challenge was class imbalance, specifically regarding the detection of the "Hazardous" class. All models exhibited lower recall for this minority category, though XGBoost detected the highest proportion of dangerous cases with a recall of 0.8712. This difficulty in detecting severe pollution cases highlights the need for further techniques such as cost-sensitive learning prior to real-world deployment. Ultimately, XGBoost was selected and saved as the final model for this task.

## 1.4 Handling Ordinality

*Does encoding ordinality via Regressor-based Ensembles outperform standard Multi-class Classifiers in Air Quality Assessment?*

Our target variable — Good, Moderate, Poor, Hazardous — is not a nominal class. It carries an inherent order:

$$\text{Good} < \text{Moderate} < \text{Poor} < \text{Hazardous}$$

Standard multi-class classification ignores this structure entirely, treating a prediction of Good when the true label is Hazardous as an error equivalent in severity to predicting Moderate. In a real-world air quality context, these are fundamentally different mistakes.

| Approach | Description |
|---|---|
| **Standard Multi-class** | Treats the 4 classes as independent nominal labels. No ordering assumed. |
| **Frank & Hall** | Decomposes the problem into 3 binary classifiers: $P(y >$ Good), $P(y >$ Moderate), $P(y >$ Poor). Outputs are combined to recover the 4-class distribution while preserving ordinal structure. |
| **Regressor Ensemble** | Trains an XGBoost Regressor on the numeric labels $\{0, 1, 2, 3\}$, then maps the continuous output back to the nearest class via tuned thresholds. |

All three models are fine-tuned with Optuna + MedianPruner. Evaluation uses both standard metrics (Accuracy, F1, MCC) and ordinal-aware metrics that penalise predictions proportionally to how far off they are.

### 1.4.1 What the metrics tell

The comparison uses both standard metrics like Accuracy, F1, and MCC, and ordinal-aware metrics that measure the magnitude of errors, not just not just if a prediction is predicting the right or wrong value

The key metrics to focus on are:

- **QWK** (Quadratic Weighted Kappa): the primary ordinal metric; higher values mean errors are concentrated near the diagonal.
- **MAE**: lower is better; tells you on average how many steps away predictions land from the true label.
- **Severe Error Rate**: the most safety-critical metric; a `Good` predicted as `Hazardous` is a worse mistake than a `Good` predicted as `Moderate`.
- **Adjacent Accuracy**: an ordinal alternative to accuracy. A prediction that is off by one class still scores.

### 1.4.2 What the results show

All three models achieve extremely high overall accuracy (between 94% and 95%) with no severe errors on the clean dataset, which reflects a synthetic nature of the data, which tells us that class boundaries are well-separated in the original feature space.

The ordinal methods do not outperform the standard classifier by a large margin on clean data, which is mainly what we expected. Because classes are easily separable, there is little ordinal structure to exploit.

### 1.4.3 Frank & Hall

The monotonicity violation rate is just 0.10%, which means the binary classifiers are producing consistent probabilities almost every time. The ordinal decomposition assumption holds well on this data, so the enforcement step is rarely needed.

### 1.4.4 Regressor Ensemble

The regressor takes a different approach, instead of decomposing the problem into binary decisions, it learns a continuous pollution score and uses Optuna-tuned thresholds to map it back to classes. This makes it more flexible and less dependent.

### 1.4.5 Conclusion

On clean data the three models are essentially tied, but the **Frank & Hall** being a little bit better. The question is what would happen if the data gets harder, if we have more examples or if the values were more overlapped. Would ordinal methods hold up better under class overlap? That is what we have done in the next part of the notebook.

### 1.4.6 Robustness Analysis

To investigate the effect of class overlap, isotropic Gaussian noise at varying levels $\sigma \in \{0.0, 1.0, 1.5, 3.0\}$ is injected into the feature space, and all three models are re-evaluated at each noise level.

### 1.4.7 Results across noise levels

As we can see in the results, as noise increases all three models perform worse, dropping from 94-95% accuracy at $\sigma = 0$ down to 83–85% at $\sigma = 3.0$. That part is expected and nearly the same across all the models and different noise levels. The more interesting differences show up in the ordinal metrics.

At $\sigma = 1.0$, all three models still produce zero severe errors, and performance is very close to the noise-less models QWK around 0.95 with Standard Multiclass slightly ahead (0.9552 vs 0.9539 and 0.9534). No method is clearly better at this noise level, as it is the most close one with the original data.

At $\sigma = 1.5$, Frank & Hall starts to be the best at every metric, but not really ahead of the other models. The ordinal models are the only ones with zero severe errors, and F&H leads on every ordinal-based metric. This seems to be the noise level where encoding ordinality starts to affect the results, but without a clear winner.

At $\sigma = 3.0$, as we expected, all the metrics for the 3 models are worse, but still quite good. All three models produce the same severe error rate, so the ordinal advantage disappears under heavy noise. Frank & Hall still leads slightly on QWK and MAE, but the gaps are still small. The Regressor Ensemble is the weakest model of the three at this high noise level.

### 1.4.8 Conclusions

To summarise, on clean or little noisy data the three models are essentially almost the same, as they produce really high, but still similar results. At moderate noise Frank & Hall is the most conservative, it avoids severe errors longer than the standard classifier and the regressor. At very high noise all methods converge and the ordinal advantage goes away. The Regressor Ensemble, despite being, in theory, the most flexible, does not outperform the other two under noise on this dataset. We also expected the results to be worse as we increased the noise level, but it hasn't been a clear winner between the

three models. As a last conclusion, we can say that both three models have learnt the mathematical formula behind the classification of the air quality level, so there is not a clear outcome of which method is a better approach to this problem as well as there isn't a clear finding on if taking into account the ordinality matters.

### 1.4.9 Future Work

The following directions could deepen the analysis:

- **Using real-world data**: validate on a real air quality dataset where class overlap exists naturally rather than being artificially injected, with real outliers, missing values, and realistic sensor characteristics.
- **Alternative ordinal methods**: explore frameworks with stronger theoretical guarantees such as CORAL or proportional odds models as additional baselines.
- **Cost-sensitive training**: explicitly penalise severe errors during training via a custom loss function, rather than only measuring them after all the process.
- **Threshold refinement**: investigate per-class or per-region threshold tuning for the Regressor Ensemble, since the optimal separation between adjacent classes may not be uniform across the feature space.
- **Realistic noise**: replace isotropic Gaussian noise with feature-specific perturbations that reflect actual sensor degradation or miscalculation patterns.

Ultimately, explicitly modelling the ordinal nature of air quality labels via the Frank & Hall method yielded the most robust and safety-conscious predictive performance.

## 1.5 Imbalanced Learning

*Does the variance-reduction of parallel ensembles offer better generalization on noisy sensor data compared to the bias-reduction of sequential ensembles?*

### 1.5.1 Managing Class Imbalance

An initial evaluation of the dataset's distribution confirmed a significant class imbalance. The critical "Hazardous" category was heavily underrepresented, making up only 10% of the data. In contrast, the majority classes dominated the dataset, with the "Good" category at 40%, "Moderate" at 30%, and "Poor" at 20%. This imbalance highlighted the necessity for specialized modeling approaches to ensure rare but dangerous air quality events are not ignored by the model.

### 1.5.2 Evaluation of Ensemble Methods

Standard ensemble architectures, such as a baseline Random Forest, were compared against specialized methods tailored for imbalanced data, including Balanced Random Forest and RUSBoost. Imbalanced-data approaches showed mixed results: while they generally focused more on the minority class, aggressive undersampling in RUSBoost led to a reduction in overall model performance. Similarly, SMOTE-Boost struggled to maintain high accuracy on the majority classes. Ultimately, the Balanced Random Forest

emerged as the strongest model in this category for accurately detecting the rare "Hazardous" cases without sacrificing too much generalized performance.

### 1.5.3 Algorithmic Modifications

To further address the imbalance, a parallel approach to data resampling was tested using algorithmic modifications via cost-sensitive learning. This involved applying severity-based penalties for misclassifications, heavily penalizing the model for missing severe air quality events. The cost-sensitive Random Forest made good use of these penalties, successfully boosting its "Hazardous" detection rates. However, despite these improvements, it still remained behind the Balanced Random Forest in overall effectiveness.

### 1.5.4 Robustness Testing for Missing Data

To simulate realistic real-world sensor failure conditions, a 30% sensor dropout was introduced to evaluate how well the algorithms could handle missing values (NaNs). In this scenario, XGBoost showcased a massive advantage due to its native capacity to handle missing data. Even with the 30% data dropout, XGBoost maintained a recall rate for the "Hazardous" class above 90%, significantly outperforming standard Random Forest pipelines that relied on traditional data imputation strategies.

### 1.5.5 Conclusion and key insights

The research concludes that Balanced Random Forest and XGBoost stand out as ideal, complementary solutions for real-world air quality monitoring. Balanced Random Forest excels at navigating severe class imbalances to catch rare, hazardous events. Meanwhile, XGBoost guarantees reliable performance, robustness, and generalization when realistic sensor dropouts and missing data occur. Used in tandem, these models provide the necessary safety and resilience required for functional, real-world monitoring systems where detecting rare events and handling noisy data are equally vital.

## 1.6 Semi-Supervised Learning

**Subtitle**

*Pregunta*

The primary goal of the experiment was to explore ways to reduce the time and cost associated with manually labeling a complete dataset by evaluating semi-supervised learning techniques. Specifically, the tasks included:

- **Simulating Unlabeled Data:** We partially deleted the target feature (Air Quality) in varying proportions across classes to mimic a real-world scenario where only a limited labeling budget is available.
- **Comparing Masking Strategies:** We implemented two different strategies for leaving data labeled:
- **Passive Semi-Supervision:** Pseudo-randomly masking target samples while entirely ignoring the class distribution.

- **Active Semi-Supervision:** Masking samples in a way that leaves the same amount of unmasked (labeled) samples from each class.
- **Evaluating Models:** We ran active and passive semi-supervised learning processes using models like Label Spreading and Self-Training (with a Random Forest base classifier). The unlabeled samples were treated as the test set to check how well the models could "self-train" and predict the missing labels compared to a supervised baseline.

The experiment yielded several key insights regarding the effectiveness of these semi-supervised methods:

- **Self-training was consistently the strongest approach** comparing it to label spreading across all ratios, specially when labeled data was low (1 to 10%), reaching the best accuracy and macro F1 in almost every train/test distribution.
- **Passive masking usually matched or outperformed active masking** for self training, this tells us that mantaining the original proportions in labeled samples worked well for our problem.
- **Active masking helped label spreading in low labeled sample proportion sets**, but this advantage was not mantained as the labeled sample ammount increased, still underperforming against self training.
- **More labeled data generally improved all methods**, but we need to take into account that those cases leave us with smaller test sets, with less samples to miss, and probably the majority of the set being part of the majority class, but good scores in macro f1 tell us that if a test sample was part of the minority class, it was correclty predicted too, therefore it worked well overall, even if we should take those metrics carefully.

If the labeling budget is limited, using just 30% to 50% labeled data already provides strong model performance. Consequently, full dataset manual labeling can be avoided with only a minimal loss in accuracy and F1 score.

## 1.7 Efficiency vs. Accuracy in Edge Use Cases

*Can lightweight ensembles replace deep stacking for edge-based Air Quality Monitoring?*

### 1.7.1 Trade-offs between Computational Cost and Classification Performance

The primary goal of the experiment was to evaluate the trade-offs between computational cost, model size, and classification performance in the context of "edge machine learning." Because air quality monitoring is typically performed on small, low-power edge devices in the city, deploying a massive ensemble model could be impractical for battery-powered chips.

The tasks included:

- **Comparing Heavy vs. Light Models:** We aimed to find the frontier between model size and performance by comparing a "Heavy Model" (a 3-layer Stacking Classifier with XGBoost, Random Forest, and SVM) against a "Light Model" (a highly pruned Decision Tree).

- **Measuring Operational Metrics:** Rather than relying solely on traditional metrics like Accuracy and F1-Score, we explicitly measured Inference Time (in milliseconds) and Model Size (in Megabytes), as they are values that have real weight when designing an edge AI soution.
- **Addressing Uncertainty (Conformal Prediction):** Since predicting air quality incorrectly (e.g., classifying "Hazardous" air as merely "Moderate") can lead to serious health problems, the notebook evaluated how certain the model was about its predictions. Using MAPIE (Model Agnostic Prediction Interval Estimator), we dealt with critical errors by computing coverage and average prediction set sizes for the different classes.

The experiment provided crucial insights into both the deployment viability and the reliability of the models:

- **Efficiency vs. Accuracy Trade-off:** While the Stacking Classifier (Heavy Model) achieved higher Accuracy (˜0.95) and Macro F1-scores (˜0.93) than the optimized Decision Tree (˜0.91 Accuracy, ˜0.87 Macro F1), the Heavy Model was severely penalized in operational cost. It was over 826 times larger in file size (MB) and more than 120 times slower overall, during inference time compared to the lightweight Decision Tree.
- **Dangerous Critical Class Coverage:** During the conformal prediction analysis, the model exhibited less than 0.90 coverage on the critical "Hazardous" class. Specifically, the true label was missing from the prediction set roughly 32% of the time, which is considered a dangerous failure rate for the most health-critical class.
- **Empty Prediction Sets:** The analysis revealed an average set size of less than 1 (or 100%), meaning that some samples were receiving empty prediction sets where no class was included at all, an anomaly that should never occur in a real world scenario.

### 1.7.2 Conclusions and Key Insights

As we expected, the heavy model took more time to draw a conclusion comparing it to the decision tree, but the target class is not correctly predicted by the smaller model in the specific sample randomly selected to test, even if it only misses the grade of the air toxicity by 1. In a real world scenario, where the minority class is the most dangerous one, missclassifying it, even if only by one space difference, can cause enormous damages, even if the results are impresive given the size difference of 17kB against 13MB. Depending in the hardware, the smaller model could do the job, but if the use case required it, the heavy model would bring more accurate results, given we optimized the hyperparameters to take the minority class more into account (with the Macro F1 Score focused optuna).

In the case of joining classes in the predictions by conformal prediction techniques, the models demonstrated good softmax values in the majority classes, but failed to output reliable values for the minority classes (where the system was previously thought to potentially benefit from joining classes).

Therefore, even if the models had small issues when predicting the minority class, the misclassifications being just 1 class appart from the true value, we believe that these are

great results given the weights and the inference time of the models, heavy and small, and depending on the hardware used for deployment, we could select the most suitable one.

# 2 Limitations and Discovery

## 2.1 Limitations and Data Drift

# 3 Ethical and Social Implications

## 3.1 Risk Assesment and Public Health

# 4 Summary and Next Steps